

# RankedShares

Proportional participatory budgeting in stablecoins. Every group of voters funds its share.

## THE PROBLEM

# 51% of the votes takes 100% of the money.

Voters

51% majority · 49% everyone else



Treasury, after a majority vote

100% to the majority's projects



In token voting a bare majority wins every proposal. The other 49% pay in, and get nothing out. This is the 51% attack on DAO treasuries, and it needs no exploit. Just a vote.

# RankedShares: every group gets its share.

1

## Money is weight

Stablecoins deposited into the pool are the budget and the voting power. Sponsors can also give weight to Uniswap liquidity providers.

Treasury, after a RankedShares round

2

## Rank the projects

Voters rank the projects they want, ties allowed. One ballot per address, applied to all of its weight. Ballots can be sealed.

3

## PB-EAR on-chain

The Expanding Approvals Rule of Aziz and Lee (AAAI 2021), tallied and verified by the pool contract. Each group's share of the weight funds its own projects.

51% one camp · 49% the other



# Fund. Propose. Vote. Tally. Claim.

## 1 · FUND THE POOL

### Money becomes weight

Depositors add USDC or EURC. Liquidity providers register their positions. Every unit of money is a unit of voting weight.

Arc · Uniswap v4

## 2 · PROPOSE

### A cost in stablecoins

Projects submit a description and a concrete amount. The organiser accepts the list and opens the round.

Arc

## 3 · VOTE

### Rank, then seal

Voters rank projects, ties allowed, and encrypt the ballot in the browser. Nobody, including the organiser, can read it.

Browser

## 4 · KEEP WEIGHT HONEST

### Prices refresh while voting

Every two minutes a workflow updates the reference price, so liquidity weight tracks value for the whole period.

Chainlink CRE

## 5 · TALLY

### Decrypt in an enclave

A confidential workflow opens the ballots, runs PB-EAR and attests the result to the pool.

Chainlink CRE

## 6 · CLAIM

### Straight from the contract

Funded projects claim their stablecoins directly. Unspent weight stays in the pool for the next round.

Arc

# Two ways to earn a say: put money in, or back the token.

The pool keeps one invariant: total voting weight equals the money in the pool. Weight can come from a deposit or from liquidity.

## DEPOSIT STABLECOINS

### One USDC is one unit of weight

Depositors add USDC or EURC and there is nothing to convert. On Arc gas is USDC too, so the budget, the weight, the costs and the fees are all one asset from the first proposal to the last claim.

## REGISTER LIQUIDITY

### Value multiplied by time

Whoever pairs the DAO token with stablecoins carries the risk first: bad spending lands on them as impermanent loss. A sponsor can give them weight. The Uniswap v4 position stays in the wallet, both legs are valued in the sponsor's stablecoin, and weight accrues for as long as the risk is carried.

# A budget is a number in a currency. Proposals need a stable one.

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## A concrete cost

A proposal asks for an amount of USDC or EURC, and that amount means the same thing on the day it is written and on the day it is paid.

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## A fixed threshold

A project is funded the moment its supporters' unspent weight reaches its cost. With a volatile token that threshold would move every block.

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## An accepted list

The organiser reviews the proposals, accepts the list and opens the round. From here on the rules do the rest.

Participatory budgeting only works when the budget holds its value from the first proposal to the last claim. That is why the pool lives on Arc.

# Ballots stay sealed. Weight stays honest.

Two things must hold for the whole voting period: nobody can read a ballot, and every unit of weight is still worth what it claims.

## SEALED BALLOTS

### Encrypted in the browser

Each voter ranks the projects, ties allowed, and encrypts the ranking to the round's public key. One ballot per address, applied to all of its weight. The ballot is opaque on-chain and off, to everyone, including the organiser.

## LIVE PRICES

### A weight fixed on day one is wrong by day seven

A Chainlink CRE workflow reads the pool price from a finalized block every two minutes and delivers a signed report. The module settles accrued credit at the old rate, then switches. Stale or out-of-order reports are refused.

# Nobody sees how you voted. The result is still correct.

Only a confidential workflow can open the ballots, and only the attested result leaves it. The pool checks the report against its own commitment before it finalizes.

## DERIVE

### Key never leaves the enclave

Inside the TEE the round's secret is derived from a master secret and the pool's public salt. Never stored, never exported.

## TALLY

### PB-EAR on plaintext

Ballots are decrypted and the full Expanding Approvals transcript runs in confidential compute.

## ATTEST

### One tuple back

The report carries the inputs root, the funded set and the transcript. The pool checks the forwarder, the workflow and its commitment.

## CLAIM

### No treasurer in between

Funded projects claim their stablecoins from the contract. Unspent weight stays in the pool for the next round.

The enclave runs as a Chainlink CRE workflow. The same pool also accepts Noir proofs and a Zisk zkVM proof of the tally: three interchangeable verifiers, one ballot format.

## THREE VERIFIERS, ONE POOL

# The same sealed ballots, three ways to trust the tally.

|                   | Chainlink CRE<br>attested                                          | Noir<br>chained proofs                                                         | ZisK<br>one proof                                                                |
|-------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| WHO TALLIES       | The Chainlink DON, in an AWS Nitro enclave.                        | The enclave writes the transcript; Noir circuits replay it in the browser.     | A ZisK guest program on the operator's machine.                                  |
| FINALITY          | One attested report through the forwarder.                         | A chain of UltraHonk proofs over the transcript, at most twelve per round.     | One PLONK proof of the whole tally.                                              |
| TRUST MODEL       | DON attestation, checked against the pool's own inputs commitment. | Sealed half proven; public half attested by the DON and auditable by anyone.   | Everything proven. No DON; the operator can only delay the result.               |
| VOTERS            | Thousands. Bounded by close gas and report size, not by the tally. | 256 sealed voters at the default profile, direct voters unbounded.             | Bounded by proving time: 2,200 voters is 15.6 M guest steps. Ceiling still open. |
| PROVING COST      | Seconds in the enclave. No proof.                                  | 1 to 9 s per proof natively, minutes per proof in a browser tab.               | About 30 min on a 32 GB CPU box: a STARK, then a PLONK wrap.                     |
| VERIFIER GAS      | One report, no cryptographic verifier.                             | 0.7 to 1.1 M gas per proof, up to twelve times.                                | 667 k gas, once.                                                                 |
| SETUP AND LEAKAGE | None. Only the result leaves the enclave.                          | No setup. The transcript reveals the sealed weight behind each funded project. | ZisK PLONK key. Nothing beyond the result.                                       |

# One stablecoin: budget, weight and gas.

[github.com/sembrestels/ranked-shares](https://github.com/sembrestels/ranked-shares)

[DEMO URL]

